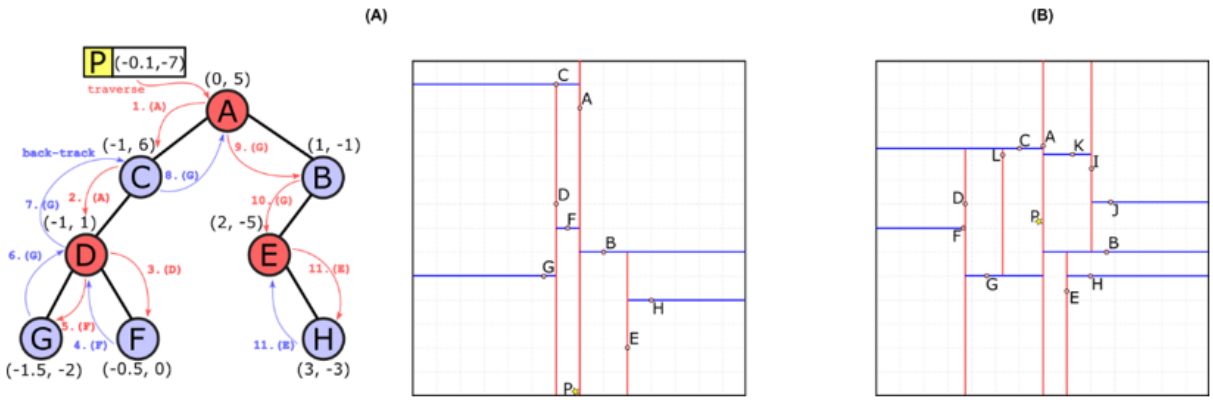


Figure 9.23 Edge cases for nearest neighbor search, which require traversing the whole tree.
(A) An example built on the k-d tree from figures 9.20-9.22: by carefully choosing the target point, we can force the algorithm to search the whole tree – as shown on the tree representation.
(B) We show the spatial representation only of an edge case where all the points lie in a circle, and we choose the center of the circle as target of the search. Notice how the distance from `P` to a split line will always be shorter than the radius of the circle (which, in turn, is the distance to the nearest neighbor).

So, there isn't unfortunately much to do: the worst-case running time for this kind of query is `O(n)`.

That's for the bad news: luckily, there is also a silver lining…

Turns out that the average running time for nearest neighbor search on a balanced k-d tree is `O(2k + log(n))`. The proof for this probabilistic bound is particularly complex and would require too much space to properly cover it here – you can find it in the original paper by Jon Bentley where k-d trees were first introduced.